

## **BRITISH STANDARDS (BS), CODE OF PRACTICE (CP) AND AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)**

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The British Standards, Code of Practice and some other international standards have hundred years foundation in construction industry and are still being currently quoted for worldwide engineering projects. These standards list out the comprehensive requirements, testing procedure and acceptance criteria for each kind of work. The young engineers and site supervisors should be well aware of these standards in relation to their trade of work by studying the requirements from time to time when they meet the terms in the drawings and specifications. A set of the commonly used Standards should be kept in project office library for reference.

Afterall, "everyone is on the same boat". To achieve a construction contract successfully with profits, necessary running costs are unavoidable and the staff's overhead is one of the main issues. As the rule "the right man on the right job" applies, the range of work found in today's projects is however so wide that the employment of experienced expertise, professional staff and qualified workers is often essential for the running of a successful project.

The following international engineering and testing standards are commonly quoted in engineering specifications for both civil engineering and building contracts . The young engineers should be well aware of these standards in relation to their trade of work by studying the requirements from time to time when they meet the terms in the drawings and specifications .

Notes : In general , the acceptance criteria of these international standards are quite conservative and with high factor of safety . Some contractor staff even say that if every site activity is going to comply with the requirements , they have no chance to complete the work on time . But more interesting is , some experienced engineers or supervisors can make cross reference of these standards for substantiation to overcome the immediate problems on site .

### **1. British Standards ( BS )**

BS 4 Structural steel sections

BS 4 : Part 1 : 1980 Specification for hot-rolled sections

BS 12 : 1978 Specification for ordinary and rapid-hardening Portland cement

BS 12 : 1989 Specification for Portland cements

BS 21 : 1985 Specification for pipe threads for tubes and fittings where pressure-tight joints are made on the threads ( metric dimensions )

BS 29 : 1976(1987) Specification for carbon steel forgings above 150 mm ruling section

BS 65 : 1988 Specification for vitrified clay pipes , fittings , joints and ducts

BS 144 : 1990 Wood preservation using coal tar creosotes

BS 373 : 1957(1986) Methods of testing small clear specimens of timber

BS 381C : 1988 Specification for colours for identification , coding and special purposes

BS 416 : 1990 Discharge and ventilating pipes and fittings , sand-cast or spun in cast iron

BS 417 : Part 2 : 1987 Specification for galvanized low carbon steel cisterns , cistern lids , tanks and cylinders - metric units

BS 427 : 1990 Method for Vickers hardness test and for verification of Vickers hardness testing machines

BS 434 : 1984 Bitumen road emulsions ( anionic and cationic )

BS 434 : Part 1 : 1984 Specification for bitumen road emulsions

BS 434 : Part 2 : 1984 Code of practice for use of bitumen road emulsions

BS 443 : 1982(1990) Specification for testing zinc coatings on steel wire and for quality requirements

BS 449 : Part 2 : 1969 Specification for the use of structural steel in building - metric units

BS 534 : 1990 Specification for steel pipes , joints and specials for water and sewage

BS 544 : 1969 Specification for linseed oil putty for use in wooden frames

BS 718 : 1979(1985) Specification for density hydrometers

BS 729 : 1971(1986) Specification for hot dip galvanized coatings on iron and steel articles

BS 743 : 1970 Specification for materials for damp-proof courses

BS 747 : 1977(1986) Specification for roofing felts

BS 812 Testing aggregates

BS 812 : Part 1 : 1975 Methods for determination of particle size and shape

BS 812 : Part 2 : 1975 Methods for determination of physical properties

BS 812 : Part 4 : 1976 Chemical properties

BS 812 : Part 101 : 1984 Guide to sampling and testing aggregates

BS 812 : Part 102 : 1989 Methods for sampling

BS 812 : Part 103 Methods for determination of particle size distribution

BS 812 : Section 103.1 : 1985 Sieve tests

BS 812 : Section 103.2 : 1989 Sedimentation test

BS 812 : Part 105 Methods for determination of particle shape

BS 812 : Section 105.1 : 1989 Flakiness index

BS 812 : Section 105.2 : 1990 Elongation index of coarse aggregate

BS 812 : Part 110 : 1990 Methods for determination of aggregate crushing value ( ACV )

BS 812 : Part 111 : 1990 Methods for determination of ten per cent fines value ( TFV )

BS 812 : Part 112 : 1990 Methods for determination of aggregate impact value ( AIV )

BS 812 : Part 113 : 1990 Methods for determination of aggregate abrasion value ( AAV )

BS 864 : Part 2 : 1983 Specification for capillary and compression fittings for copper tubes

BS 873 Road traffic signs and internally illuminated bollards

BS 873 : Part 1 : 1983 Methods of test

BS 873 : Part 5 : 1983 Specification for internally illuminated signs and external lighting luminaries

BS 873 : Part 6 : 1983 Specification for retroreflective and non-retroreflective signs

BS 873 : Part 7 : 1984 Specification for posts and fittings

BS 882 : 1983 Specification for aggregates from natural sources for concrete

BS 890 : 1972 Specification for building limes

BS 903 Physical testing of rubber

BS 903 : Part A1 : 1980(1988) Determination of density

BS 903 : Part A2 : 1989 Determination of tensile stress-strain properties

BS 903 : Part A3 : 1982 Determination of tear strength ( trouser , angle and crescent test pieces )

BS 903 : Part A4 : 1990 Determination of compression stress-strain properties

BS 903 : Part A5 : 1974 Determination of tension set

BS 903 : Part A6 : 1969 Determination of compression set after constant strain

BS 903 : Part A9 : 1988 Determination of abrasion resistance

BS 903 : Part A16 : 1987 Determination of the effect of liquids

BS 903 : Part A18 : 1973(1985) Determination of equilibrium water vapour absorption

BS 903 : Part A19 : 1986 Heat resistance and accelerated ageing tests

BS 903 : Part A26 : 1969 Determination of hardness

BS 903 : Part A43 : 1990 Method for determination of resistance to ozone cracking ( static strain test )

BS 903 : Part C2 : 1982 Determination of volume resistivity

BS 952 Glass for glazing

BS 952 : Part 1 : 1978 Classification

BS 952 : Part 2 : 1980 Terminology for work on glass

BS 970 : Part 1 : 1983 General inspection and testing procedures and specific requirements for carbon , carbon manganese , alloy and stainless steels

BS 1004 : 1972(1985) Specification for zinc alloys for die casting and zinc alloy die casting

BS 1010 : Part 2 : 1973 Draw-off taps and above-ground stopvalves

BS 1014 : 1975(1986) Specification for pigments for Portland cement and Portland cement products

BS 1070 : 1973(1979) Specification for black paint (tar-based)

BS 1155 : 1986 Specification for natural rubber compounds  
for extrusion

BS 1611 : 1977(1984) Specification for aluminium alloy sections for structural purposes

BS 1181 : 1989 Specification for clay flue linings and flue terminals

BS 1191 Specification for gypsum building plasters

BS 1191 : Part 1 : 1973 Excluding premixed lightweight plasters

BS 1194 : 1969 Specification for concrete porous pipes for under-drainage

BS 1199 and 1200 : 1976 Specification for building sands from natural sources

BS 1203 : 1979 Specification for synthetic resin adhesives (phenolic and aminoplastic) for plywood

BS 1204 Synthetic resin adhesives (phenolic and aminoplastic) for wood

BS 1204 : Part 1 : 1979(1991) Specification for gap-filling adhesives

BS 1212 Float operated valves

BS 1212 : Part 1 : 1990 Specification for piston type float operated valves (copper alloy body) (excluding floats)

BS 1212 : Part 2 : 1990 Specification for diaphragm type float operated valves (copper alloy body) (excluding floats)

BS 1212 : Part 3 : 1990 Specification for diaphragm type float operated valves (plastic bodied) for cold water services only (excluding floats)

BS 1247 : 1990 Manhole steps

BS 1336 : 1971(1988) Specification for knotting

BS 1369 Steel lathing for internal plastering and external rendering

BS 1369 : Part 1 : 1987 Specification for expanded metal and ribbed lathing

BS 1377 : 1975 Methods of test for soils for civil engineering purposes

BS 1387 : 1985(1990) Specification for screwed and socketed steel tubes and tubulars and for plain end steel tubes suitable for welding or for screwing to BS 21 pipe threads

BS 1400 : 1985 Specification for copper alloy ingots and copper alloy and high conductivity copper castings

BS 1449 : Part 1 : 1983 Specification for carbon and carbon-manganese plate , sheet and strip

BS 1449 : Part 2 : 1983 Specification for stainless and heat-resisting steel plate , sheet and strip

BS 1452 : 1990 Specification for flake graphite cast iron

BS 1470 : 1987 Specification for wrought aluminium and aluminium alloys for general engineering purposes : plate , sheet and strip

BS 1471 : 1972 Specification for wrought aluminium and aluminium alloys for general engineering purposes – drawn tube

BS 1473 : 1972 Specification for wrought aluminium and aluminium alloys for general engineering purposes – rivet , bolt and screw stock

BS 1474 : 1987 Specification for wrought aluminium and aluminium alloys for general engineering purposes : bars , extruded round tubes and sections

BS 1494 : Part 2 : 1967 Sundry fixings

BS 1610 : 1985 Materials testing machines and force verification equipment

BS 1615 : 1987 Method for specifying anodic oxidation coatings on aluminium and its alloys

BS 1722 : Part 1 : 1986 Specification for chain link fences

BS 1740 : Part 1 : 1971(1990) Specification for wrought steel pipe fittings (screwed BS 21 R-series thread)

BS 1924 : 1990 Stabilized materials for civil engineering purposes

BS 2000 Methods of test for petroleum and its products

BS 2015 : 1965(1985) Glossary of paint terms

BS 2451 : 1963(1988) Specification for chilled iron shot and grit

BS 2456 : 1990 Specification for floats (plastics) for float operated valves for cold water services

BS 2494 : 1990 Specification for elastomeric seals for joints in pipework and pipelines

BS 2499 : 1973 Specification for hot applied joint sealants for concrete pavements

BS 2523 : 1966(1983) Specification for lead-based priming paints

BS 2569 : Part 1 : 1964(1988) Protection of iron and steel by aluminium and zinc against atmospheric corrosion

BS 2600 Radiographic examination of fusion welded butt joints in steel

BS 2600 : Part 1 : 1983 Methods for steel 2 mm up to and including 50 mm thick

BS 2600 : Part 2 : 1973 Methods for steel over 50 mm up to and including 200 mm thick

BS 2633 : 1987 Specification for Class I arc welding for ferritic steel pipework for carrying fluids

BS 2648 : 1955 Performance requirements for electrically- heated laboratory drying ovens

BS 2760 : 1973 Specification for pitch-impregnated fibre pipes and fittings for below and above ground drainage

BS 2782 Methods of testing plastics

BS 2782 : Part 3 :

Methods 320A to 320F : 1976 Tensile strength , elongation and elastic modulus

BS 2782 : Part 3 :

Method 365A : 1976(1989) Determination of softness number of flexible plastics materials

BS 2782 : Part 3 :

Method 365D : 1978(1983) Determination of hardness of plastics and ebonite by the ball indentation method

BS 2782 : Part 4 :

Methods 430A to 430D : 1983 Determination of water absorption at 23 ° C .

Determination of water absorption at 23 ° C with allowance for water-soluble matter .

Determination of boiling water absorption .

Determination of boiling water absorption with allowance for water-soluble matter .

BS 2782 : Part 6 :

Methods 620A to 620D : 1980 Determination of density of solid plastics excluding cellular plastics (immersion method) .

Determination of density of solid plastics excluding cellular plastics (pycnometer method) .

Determination of plastics (sink-float method) .

Determination of density of solid plastics excluding cellular plastics (density gradient column method)

BS 2789 : 1985 Specification for spheroidal graphite or nodular graphite cast iron

BS 2846 : Part 3 : 1975(1985) Determination of a statistical tolerance interval

BS 2846 : Part 4 : 1976(1985) Techniques of estimation and tests relating to means and variances

BS 2869 : Part 2 : 1988 Specification for fuel oil for agricultural and industrial engines and burners (classes A2 , C1 , C2 , D , E , F , G and H)

BS 2871 : Part 1 : 1971 Copper tubes for water , gas and sanitation

BS 2874 : 1986 Specification for copper and copper alloy rods and sections (other than forging stock)

BS 2910 : 1986 Methods for radiographic examination of fusion welded circumferential butt joints in steel pipes

BS 2989 : 1982 Specification for continuously hot-dip zinc coated and iron-zinc alloy coated steel : wide strip , sheet / plate and slit wide strip

BS 3019 : Part 1 : 1984 Specification for TIG welding of aluminum , magnesium and their alloys

BS 3049 : 1976 Specification – Pedestrian guard rails (metal)

BS 3100 : 1976(1984) Specification for steel castings for general engineering purposes

BS 3148 : 1980 Methods of test for water for making concrete (including notes on the suitability of the water)

BS 3262 : Part 1 : 1989 Specification for constituent materials and mixtures

BS 3262 : Part 3 : 1989 Specification for application of material to road surfaces

BS 3382 : Part 1 & 2 : 1961 Cadmium on steel components Zinc on steel components

BS 3410 : 1961 Specification for metal washers for general engineering purposes

BS 3416 : 1988 Specification for bitumen-based coatings for cold application , suitable for use in contact with potable water

BS 3468 : 1986 Specification for austenitic cast iron

BS 3505 : 1986 Specification for unplasticized polyvinyl chloride (PVC-U) pressure pipes for cold potable water

BS 3506 : 1969 Specification for unplasticized PVC pipe for industrial uses

BS 3571 : Part 1 : 1985 Specification for MIG welding of aluminium and aluminium alloys

BS 3600 : 1976(1988) Specification for dimensions and masses per unit length of welded and seamless steel pipes and tubes for pressure purposes

BS 3601 : 1987 Specification for carbon steel pipes and tubes with specified room temperature properties for pressure purposes

BS 3690 : Part 1 : 1989 Specification for bitumens for roads and other paved areas

BS 3690 : Part 2 : 1989 Specification for bitumens for industrial purposes

BS 3692 : 1967 Specification for ISO metric precision hexagon bolts , screws and nuts. Metric units

BS 3698 : 1964(1979) Specification for calcium plumbate priming paints

BS 3892 : Part 1 : 1982 Specification for pulverized-fuel ash for use as a cementitious component in structural concrete

BS 3900 : Part E10: 1979(1989) Mechanical tests on paint films – Pull-off test for adhesion

BS 3923 : Part 1 : 1986 Methods for manual examination of fusion welds in ferritic steels

BS 3923 : Part 2 : 1972 Automatic examination of fusion welded butt joints in ferritic steels

BS 3981 : 1976(1985) Specification – Iron oxide pigments for paints

BS 3987 : 1974 Specification for anodic oxide coatings on wrought aluminum for external architectural applications

BS 3998 : 1989 Recommendations for tree work

BS 4027 : 1980 Specification for sulphate-resisting Portland cement

BS 4072 : 1987 Wood preservation by means of copper / chromium / arsenic compositions

BS 4102 : 1990 Specification for steel wire and wire products for fences

BS 4147 : 1980(1987) Specification for bitumen-based hot-applied coating materials for protecting iron and steel , including suitable primers where required

BS 4168 : Part 1 : 1981 Specification for hexagon socket head cap screws

BS 4190 : 1967 Specification for ISO metric black hexagon bolts , screws and nuts

BS 4211 : 1987 Specification for ladders for permanent access to chimneys , other high structures , silos and bins

BS 4232 : 1967 Specification for surface finish of blast-cleaned steel for painting

BS 4254 : 1983 Specification for two-part polysulphide-based sealants

BS 4320 : 1968 Specification for metal washers for general engineering purposes. Metric series

BS 4345 : 1968(1986) Specification for slotted angles

BS 4346 Joints and fittings for use with unplasticized PVC pressure pipes

BS 4346 : Part 1 : 1969 Injection moulded unplasticized PVC fittings for solvent welding for use with pressure pipes , including potable water supply

BS 4346 : Part 2 : 1970 Mechanical joints and fittings , principally of unplasticized PVC

BS 4346 : Part 3 : 1982 Specification for solvent cement

BS 4360 : 1990 Specification for weldable structural steels

BS 4393 : 1969(1985) Specification for tin or tin-lead coated copper wire

BS 4395 Specification for high strength friction grip bolts and associated nuts and washers for structural engineering

BS 4395 : Part 1 : 1969 General grade

BS 4395 : Part 2 : 1969 Higher grade bolts and nuts and general grade washers

BS 4395 : Part 3 : 1973 Higher grade bolts (waisted shank) , nuts and general grade washers

BS 4447 : 1973(1990) Specification for the performance of prestressing anchorage for post-tensioned construction

BS 4449 : 1978(1984)

( excluding AMD 4337 and AMD 4540 and clauses 12.2.1 and 12.2.2 ) Specification for hot rolled steel bars for the reinforcement of concrete

BS 4461 : 1978 Specification for cold worked steel bars for the reinforcement of concrete

BS 4466 : 1989 Specification for scheduling , dimensioning , bending and cutting of steel reinforcement for concrete

BS 4482 : 1985 Specification for cold reduced steel wire for the reinforcement of concrete

BS 4483 : 1985 Specification for steel fabric for the reinforcement of concrete

BS 4486 : 1980 Specification for hot rolled and hot rolled and processed high tensile alloy steel bars for the prestressing of concrete

BS 4504 : Section 3.1 : 1989 Circular flanges for pipes , valves and fittings ( PN designated ) – Specification for steel flanges

BS 4514 : 1983 Specification for unplasticized PVC soil and ventilating pipes , fittings and accessories

BS 4515 : 1984 Specification for welding of steel pipelines on land and offshore

BS 4550 : Part 1 : 1978 Methods of testing cement – Sampling

BS 4550 : Part 2 : 1970 Methods of testing cement – Chemical tests

BS 4550 : Part 3 : 1978 Methods of testing cement – Physical tests

BS 4551 : 1980 Methods of testing mortars , screeds and plasters

BS 4568 Specification for steel conduit and fittings with metric threads of ISO form electrical installations

BS 4568 : Part 1 : 1970 Steel conduit , bends and couplers

BS 4568 : Part 2 : 1970(1988) Fittings and components

BS 4570 : 1985 Specification for fusion welding of steel castings

BS 4576 : Part 1 : 1989 Unplasticized polyvinyl chloride ( PVC-U ) rainwater goods and accessories – Half-round gutters and pipes of circular cross-section

BS 4604 Specification for the use of high strength friction grip bolts in structural steelwork

BS 4604 : Part 1 : 1970 General grade

BS 4604 : Part 2 : 1970 Higher grade ( parallel shank )

BS 4620 : 1970(1988) Specification for rivets for general engineering purposes

BS 4622 : 1970(1983) Specification for grey iron pipes and fittings

BS 4652 : 1971 (1979) Specification for metallic zinc-rich priming paint ( organic media )

BS 4660 : 1989 Specification for unplasticized polyvinyl chloride ( PVC-U ) pipes and plastic fittings of nominal sizes 110 and 160 for below ground gravity drainage and sewerage

BS 4662 : 1970(1989) Specification for boxes for the enclosure of electrical accessories

BS 4677 : 1984 Specification for arc welding of austenitic stainless steel pipework for carrying fluids

BS 4756 : 1971(1983) Specification for ready mixed aluminium priming paints for woodwork

BS 4772 : 1988 Specification for ductile iron pipes and fittings

BS 4848 Specification for hot-rolled structural steel sections

BS 4848 : Part 2 : 1975 Hollow sections

BS 4848 : Part 4 : 1972(1986) Equal and unequal angles

BS 4848 : Part 5 : 1980 Bulb flats

BS 4865 : Part 1 : 1989 Specification for non-metallic flat gaskets ( including gaskets for flanges to BS 4772 )

BS 4870 : Part 1 : 1981 Specification for approval testing of welding procedures – Fusion welding of steel

BS 4871 : Part 1 : 1982 Specification for approval testing of welders working to approved welding procedures – Fusion welding of steel

BS 4872 : Part 1 : 1982 Specification for approval testing of welders when welding procedure approval is not required – Fusion welding of steel

BS 4873 : 1986 Specification for aluminium alloy windows

BS 4921 : 1988 Specification for sheradized coatings on iron or steel

BS 4933 : 1973 Specification for ISO metric black cup and countersunk head bolts and screws with hexagon nuts



BS 4942 : 1981 Short link chain for lifting purposes

BS 5075 : Part 1 : 1982 Specification for accelerating admixtures , retarding admixtures and water reducing admixtures

BS 5075 : Part 3 : 1985 Specification for superplasticizing admixtures

BS 5135 : 1984 Specification for arc welding of carbon and carbon manganese steels

BS 5150 : 1990 Specification for cast iron gate valves

BS 5154 : 1989 Specification for copper alloy globe , globe stop and check , check and gate valves

BS 5163 : 1986 Specification for predominantly key-operated cast iron gate valves for waterworks purposes

BS 5212 : 1990 Cold applied joint sealant systems for concrete pavements

BS 5215 : 1986 Specification for one-part gun grade polysulphide-based sealants

BS 5252F : 1976(1986) Colour matching fan

BS 5255 : 1989 Specification for thermoplastics waste pipe and fittings

BS 5268 : Part 2 : 1988 Structural use of timber – Code of practice for permissible stress design , materials and workmanship

BS 5284 : 1976 Methods , sampling and testing mastic asphalt and pitchmastic used in building

BS 5289 : 1976(1983) Code of practice – Visual inspection of fusion welded joints

BS 5395 : Part 1 : 1977(1984) Code of practice for the design of straight stairs

BS 5400 Steel , concrete and composite bridges

BS 5400 : Part 2 : 1978 Specification for loads

BS 5400 : Part 4 : 1990 Code of practice for design of concrete bridges

BS 5400 : Part 6 : 1980 Specification for materials and workmanship , steel

BS 5400 : Part 9 : 1983 Bridge bearings

BS 5400 : Section 9.2 : 1983 Specification for materials , manufacture and installation of bridge bearings

BS 5481 : 1977(1989) Specification for unplasticized PVC pipes and fittings for gravity sewers

BS 5493 : 1977 Code of practice for protective coating of iron and steel structures against corrosion

BS 5573 : 1978 Code of practice for safety precautions in the construction of large diameter boreholes for piling and other purposes

BS 5589 : 1989 Code of practice for preservation of timber

BS 5756 : 1980(1985) Specification for tropical hardwoods graded for structural use

BS 5896 : 1980 Specification for high tensile steel wire and strand for the prestressing of concrete

BS 5911 Precast concrete pipes , fittings and ancillary products

BS 5911 : Part 2 : 1982 Specification for inspection chambers and street gullies

BS 5911 : Part 3 : 1982 Specification for pipes and fittings with ogee joints

BS 5911 : Part 100 : 1988 Specification for unreinforced and reinforced pipes and fittings with flexible joints

BS 5911 : Part 200 : 1989 Specification for unreinforced and reinforced manholes and soakaways of circular cross section

BS 5930 : 1981 Code of practice for site investigations

BS 5931 : 1980 Code of practice for machine laid in situ edge details for paved area

BS 5950 : Part 2 : 1985 Specification for materials , fabrication and erection : hot rolled sections

BS 5996 : 1980 Methods for ultrasonic testing and specifying quality grades of ferritic steel plate

BS 6044 : 1987 Specification for pavement marking paints

BS 6072 : 1981(1986) Method for magnetic particle flaw detection

BS 6088 : 1981(1985) Specification for solid glass beads for use with road marking compounds and for other industrial uses

BS 6089 : 1981 Guide to assessment of concrete strength in existing structures

BS 6105 : 1981 Specification for corrosion-resistant stainless steel fasteners

BS 6150 : 1982 Code of practice for painting of buildings

BS 6262 : 1982 Code of practice for glazing for buildings

BS 6323 : 1982(1990) Specification for seamless and welded steel tubes for automobile , mechanical and general engineering purposes

BS 6323 : Part 1 : 1982(1990) General requirements

BS 6323 : Part 3 : 1982(1990) Specific requirements for hot finished seamless steel tubes

BS 6323 : Part 8 : 1982(1990) Specific requirements for longitudinally welded stainless steel tubes

BS 6349 : Part 1 : 1984 Code of practice for maritime structures – General criteria

BS 6362 : 1990 Specification for stainless steel tubes suitable for screwing in accordance with BS 21 " Pipe threads for tubes and fittings where pressure-tight joint are made on the threads "

BS 6405 : 1984 Specification for non-calibrated short link steel chain ( grade 30 ) for general engineering purposes : class 1 and 2

BS 6431 Ceramic floor and wall tiles

BS 6431 : Part 1 : 1983 Specification for classification and marking , including definitions and characteristics

BS 6431 : Part 2 : 1984 Specification for extruded ceramic tiles with a low water absorption (  $E \leq 3\%$  ). Group A1

BS 6431 : Part 3 Extruded ceramic tiles with a water absorption of  $3\% < E \leq 6\%$  . Group A11a

BS 6431 : Part 3 : Section 3.1 : 1986 Specification for general products

BS 6431 : Part 3 : Section 3.2 : 1986 Specification for products ( terre cuite , cotto , baldosin catalan )

BS 6431 : Part 4 Extruded ceramic tiles with a water absorption of  $6\% < E \leq 10\%$  . Group A11b

BS 6431 : Part 4 : Section 4.1 : 1986 Specification for general products

BS 6431 : Part 4 : Section 4.2 : 1986 Specification for specific products ( terre cuite , cotto , baldosin catalan )

BS 6443 : 1984 Method for penetrant flaw detection

BS 6463 Quicklime , hydrated lime and natural calcium carbonate

BS 6463 : Part 1 : 1984 Methods of sampling

BS 6463 : Part 2 : 1984 Methods of chemical analysis

BS 6463 : Part 4 : 1987 Methods of test for physical properties of hydrated lime and lime putty

BS 6510 : 1984 Specification for steel windows , sills , window boards and doors

BS 6558 : 1985 Optical fibres and cables

BS 6566 Plywood

BS 6577 : 1985 Specification for mastic asphalt for building ( natural rock asphalt aggregate )

BS 6588 : 1985 Specification for Portland pulverized-fuel ash cement

BS 6657 : 1986 Guide for prevention of inadvertent initiation of electro-explosive devices by radio-frequency radiation

BS 6681 : 1986 Specification for malleable cast iron

BS 6700 : 1987 Specification for design , installation , testing and maintenance of services supplying water for domestic use within buildings and their cartilages

BS 6925 : 1988 Specification for mastic asphalt for building and civil engineering ( limestone aggregate )

BS 6949 : 1988 Specification for bitumen-based coatings for cold application , excluding use in contact with potable water

BS 7263 : 1990 Precast concrete flags , kerbs , channels , edgings and quadrants

BS 7263 : Part 1 : 1990 Specification

BS 8004 : 1986 Code of practice for foundations

BS 8005 : Part 1 : 1987 Guide to new sewerage construction

## **2. Code of Practice ( CP )**

CP 144 Roof coverings

CP 144 : Part 4 : 1970 Mastic asphalt . Metric units

## **3. American Society for Testing and Materials ( ASTM ) Standards**

ASTM A 775M-86 Specification for epoxy-coated reinforcing steel bars

ASTM C 127-88 Test method for specific gravity and absorption of coarse aggregate

ASTM C 128-88 Test method for specific gravity and absorption of fine aggregate

ASTM C 131-81(1987) Test method for resistance to degradation of small-size coarse aggregate by abrasion and impact in the Los Angeles Machine

ASTM C 136-84a Method for sieve analysis of fine and coarse aggregates

ASTM C 188-84 Test method for density of hydraulic cement

ASTM C 939-87 Test method of flow of grout for preplaced- aggregate concrete

ASTM D 5-86 Test method for penetration of bituminous materials

ASTM D 113-86 Test method for ductility of bituminous materials

ASTM D 140-88 Method for sampling bituminous materials

ASTM D 242-85 Specification for mineral filler for bituminous paving mixtures

ASTM D 546-88 Method for sieve analysis of mineral filler for road and paving materials

ASTM D 854-83 Test method for specific gravity of soils

ASTM D 946-82 Specification for penetration-graded asphalt cement for use in pavement construction

ASTM D 979-87 Method for sampling bituminous paving mixtures

ASTM D 1559-82 Test method for resistance to plastic flow of bituminous mixtures using Marshall apparatus

ASTM D 1754-87 Test method for effect of heat and air on asphaltic materials ( thin-film over test )

ASTM D 2000-86 Classification system for rubber products in automobile applications

ASTM D 2027-76(1986) Specification for cutback asphalt ( medium-curing type )

ASTM D 2041-78 Test method for theoretical maximum specific gravity of bituminous paving mixtures

ASTM D 2042-81(1985) Test method for solubility of asphalt materials in trichloroethylene

ASTM D 2171-88 Test method for viscosity of asphalts by vacuum capillary

ASTM D 2172-88 Test method for quantitative extraction of bitumen from bituminous paving mixtures

ASTM D 2726-88 Test method for bulk specific gravity of compacted bituminous mixtures using saturated surface-dry specimens

ASTM D 3203-88 Test method for percent air voids in compacted dense and open bituminous paving mixtures

ASTM D 3289-85 Test method for specific gravity or density of semi-solid and solid bituminous materials by nickel crucible

